8. 1

1

(3) ✓ $$\int \frac{\sqrt{x}}{x+x\sqrt{x}}\,dx$$

$$= \int \frac{\cancel{\sqrt{x}}}{\underset{\sqrt{x}}{\cancel{x}}(1+\sqrt{x})}\,dx$$

$1+\sqrt{x} = t$

$\frac{1}{2\sqrt{x}}\,dx = dt$

$$\int \frac{1}{t}\cdot 2\,dt$$

$$= 2\cdot \ln|t| + C = 2\ln|1+\sqrt{x}| + C \text{ ″}$$

(6) ✓ $$\int \frac{1}{x\ln x}\,dx$$

$\ln x = t$

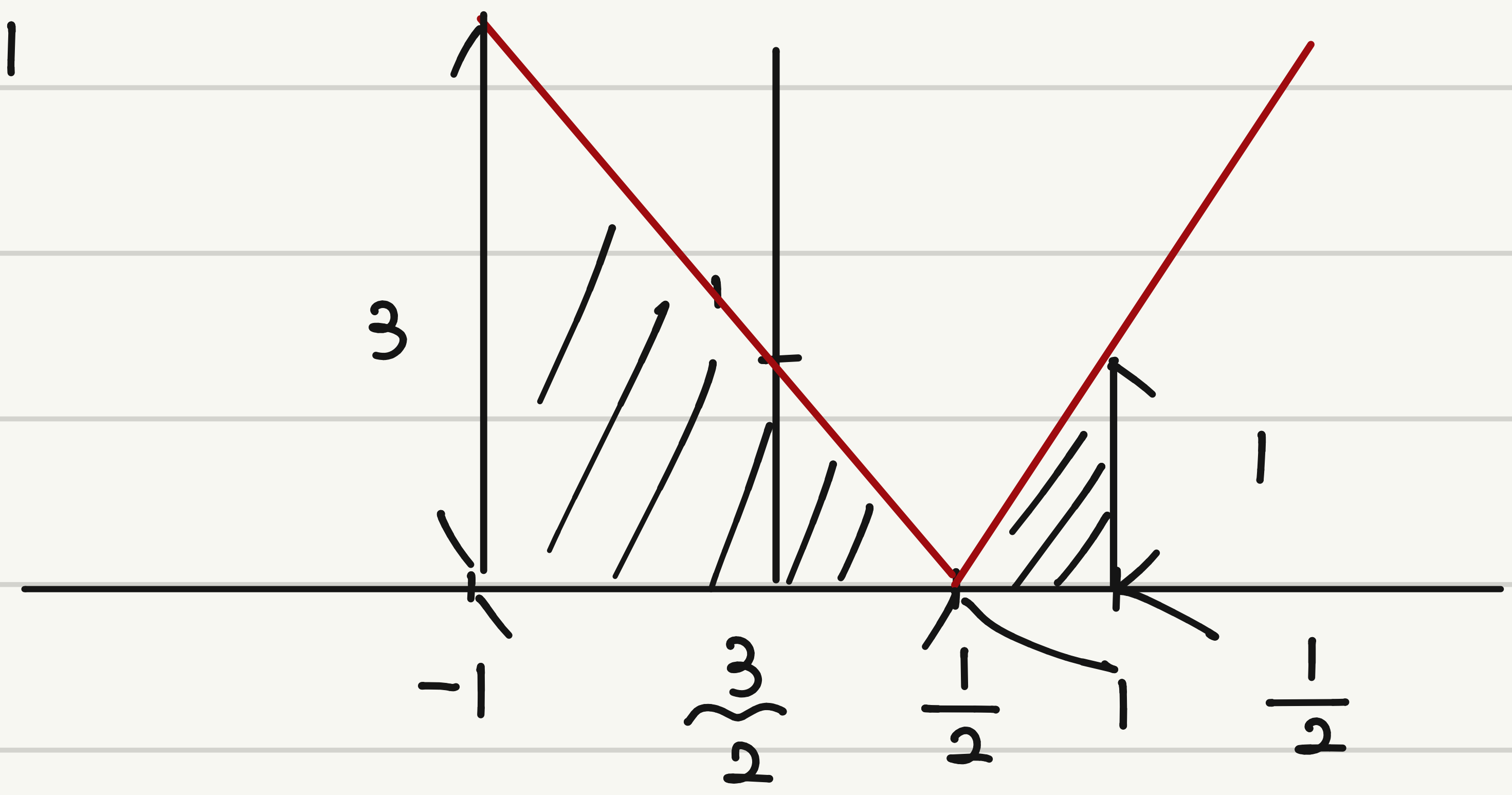
$\frac{1}{x}\,dx = dt$

$$\int \frac{1}{t}\,dt$$

$$= \ln|t| + C = \ln|\ln x| + C \text{ ″}$$

2

(2) ✓ $$\int_{-1}^{1} |2x-1|\,dx$$



$$\therefore \frac{9}{4} + \frac{1}{4} = \frac{5}{2} \text{ ″}$$

3

(2) ✓ $\int \frac{1}{x-\sqrt[3]{x}}\, dx$

$\sqrt[3]{x} = t$

$\frac{1}{3} x^{-\frac{2}{3}}\, dx = dt \quad \cdots \quad dx = 3\cdot x^{\frac{2}{3}}\, dt = 3t^2\, dt$

$\int \frac{1}{t^3-t}\cdot 3t^2\, dt$

$= \int \frac{3t}{t^2-1}\, dt$

$= \frac{3}{2}\ln|t^2-1| = \frac{3}{2}\ln|x^{\frac{2}{3}}-1|$ //

(6) ✓ $\int \frac{e^x}{(e^x+1)\ln(e^x+1)}\, dx$

$e^x+1 = t$

$e^x\, dx = dt$

$\int \frac{1}{t\ln t}\, dt$

$\ln t = u$

$\frac{1}{t}\, dt = du$

$\int \frac{1}{u}\, du$

$= \ln|u|$

$= \ln|\ln t|$

$= \ln|\ln(e^x+1)|$ //

4

$\pi - x = \theta \qquad -dx = d\theta$

$\int_0^{\pi} \theta f(\sin\theta)\, d\theta$

$= -\int_{\pi}^{0} (\pi-x) f(\sin(\pi-x))\, dx$

$= \int_0^{\pi} (\pi-x) f(\sin x)\, dx$

#4 ✓ 이어서

$$\pi\int_0^{\pi} f(\sin x)\,dx - \int_0^{\pi} x f(\sin x)\,dx$$

$$\therefore\ \pi\int_0^{\pi} f(\sin\theta)\,d\theta - \int_0^{\pi} \theta f(\sin\theta)\,d\theta = \int_0^{\pi} \theta f(\sin\theta)\,d\theta$$

$$\therefore\ \frac{\pi}{2}\int_0^{\pi} f(\sin\theta)\,d\theta = \int_0^{\pi} \theta f(\sin\theta)\,d\theta$$

#6 ✓

$$\int_0^{60} 100\cdot e^{-0.01t}\,dt$$

$$= 100\cdot\Big[-100\cdot e^{-0.01t}\Big]_0^{60}$$

$$= 100\left(-100\cdot e^{-0.6} + 100\right)$$

$$= 10000\left(1 - e^{-0.6}\right)\ ''$$

#8 ✓

(3) $$\int \frac{dx}{x\sqrt{2x-x^2}}$$

$x = \frac{1}{u}$ $\quad dx = -\frac{1}{u^2}\,du$

$$\int \frac{1}{\frac{1}{\cancel{u}}\sqrt{\frac{2}{u}-\frac{1}{u^2}}}\cdot\left(-\frac{1}{u^{\cancel{2}}}\,du\right)$$

$$= -\int \frac{1}{\sqrt{\frac{2}{u}-\frac{1}{u^2}}}\cdot\frac{1}{u}\,du$$

$$= -\int \frac{1}{\sqrt{\frac{2u-1}{u^2}}}\cdot\frac{1}{u}\,du = -\int \frac{1}{\sqrt{2u-1}}\,du$$

$$= -(2u-1)^{\frac{1}{2}} + C$$

$$= -\left(\frac{2}{x}-1\right)^{\frac{1}{2}} + C\ ''$$